

Checkpointed Newton–Nesterov (CN^2): A Stage-Wise Switching Framework for Hybrid First– and Second–Order Optimization

Jay

Collaborators

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Abstract

We introduce a stage-wise switching framework, *Checkpointed Newton–Nesterov* (CN^2), that alternates between a coarse *Nesterov accelerated gradient* (NAG) phase and a local *Newton* phase. Rather than continuously blending momentum with curvature information—as in recent inertial cubic-regularized Newton methods— CN^2 performs a *hard* switch: when the Newton decrement $\lambda(x) = \sqrt{\nabla f(x)^\top [H(x)]^{-1} \nabla f(x)}$ indicates we have left the solution basin, we navigate with cheap first-order accelerated steps; once the decrement falls below a user threshold τ , we commit to damped Newton for quadratic local convergence. In nonconvex regions where the Hessian is not positive definite, we fall back to a gradient-norm gate, making the method robust on functions such as Rosenbrock and Beale where vanilla Newton fails abruptly. On a battery of test problems, CN^2 converges on every instance—including Rosenbrock in dimension $n = 200$, where plain Newton, Newton-CG, and L-BFGS all stall—while using a comparable or smaller number of Hessian evaluations. The method is robust to the user parameter K_0 (jump length) and exposes a clean accuracy-cost trade-off in the threshold τ .

1 Introduction

Newton’s method attains quadratic local convergence but incurs an $\mathcal{O}(n^3)$ cost per iteration for the dense Hessian solve, rendering it prohibitive at scale. Nesterov accelerated gradient (NAG) methods achieve the optimal $\mathcal{O}(1/k^2)$ rate for smooth convex problems but slow down geometrically near the solution, especially on ill-conditioned objectives. This tension motivates hybrid designs that recover higher-order local behavior without paying the full second-order cost on every step.

Recent work has pursued *continuous* blending of momentum with cubic regularization, *e.g.* inertial cubic-regularized Newton (ICN) [1], accelerated cubic Newton variants [2], and momentum-accelerated stochastic cubic methods [3]. These methods intertwine momentum and curvature on a per-step basis. In contrast, this paper studies a sharper, arguably more interpretable design: we keep the two algorithmic families *separate* and switch between them only at coarse *checkpoints*. This is the “coarse-to-fine navigation” philosophy: expensive Newton assessments are performed rarely, and accelerated gradient steps fill the gaps.

Our contribution is a principled *stage-wise* method that:

1. runs a short Newton burst (two damped steps) and evaluates the Newton decrement;
2. if the decrement exceeds a threshold (far from the solution), runs a Nesterov phase until the decrement—measured with a rare, dimension-adaptive Hessian cadence—enters the solution basin;

3. falls back to a gradient-norm gate in nonconvex regions where the Hessian is not positive definite, ensuring global robustness.

2 Background and Related Work

Newton and trust regions. Damped Newton with backtracking retains global convergence on convex objectives subject to Lipschitz-continuous Hessians. Newton–CG [6] avoids forming H by solving the Newton system with conjugate gradients. Quasi-Newton methods such as L-BFGS approximate curvature from gradient differences and need no Hessian at all.

Nesterov acceleration. The accelerated gradient method [4] reaches $f(x_k) - f^* = \mathcal{O}(1/k^2)$ for L -smooth convex functions. It is attractive when evaluations of the gradient are cheap relative to second-order information.

Cubic regularization. Trust-region and cubic-regularized Newton (CRN) [6, 5] restore global convergence while achieving $\mathcal{O}(1/k^{3/2})$ (convex) and superlinear/quadratic rates near the solution, at the price of a subproblem solve involving the Hessian.

Recent hybridizations (2025–2026). Inertial cubic-regularized Newton (ICN) [1] explicitly fuses NAG-style momentum with the cubic framework; OPTAMI/NATA [2] provide accelerated tensor methods; and momentum-accelerated stochastic cubic variants [3] tackle the stochastic regime. Our work differs by keeping the phases *disjoint* and switching on a *geometric* measure (the Newton decrement), rather than blending operators.

3 The CN^2 Algorithm

3.1 Setup and notation

We minimize a smooth (possibly nonconvex) function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Let ∇f denote the gradient and $H = \nabla^2 f$ the Hessian. The *Newton decrement* is

$$\lambda(x) = \sqrt{\nabla f(x)^\top (H(x) + \rho I)^{-1} \nabla f(x)}, \quad (1)$$

where $\rho \geq 0$ is a small regularization shift. For convex objectives with $H \succ 0$, $\lambda(x)$ is a natural proximity measure to the solution.

3.2 Dual entry gate

When H is not positive definite (nonconvex region), the decrement (1) is not meaningful. We therefore define the entry measure

$$g(x) = \begin{cases} \lambda(x), & H(x) \succ 0, \\ \|\nabla f(x)\|, & \text{otherwise,} \end{cases} \quad (2)$$

and compare it against a threshold τ (for the Hessian case) or τ_g (for the gradient case). This dual gate (option “c”) is what makes the method robust on nonconvex test functions such as Rosenbrock and Beale.

Algorithm 1 CN^2 (Checkpointed Newton–Nesterov)

- 1: Given x_0 , thresholds $\tau, \tau_g > 0$, jump length K_0 , Newton steps $m = 2$.
 - 2: **repeat**
 - 3: **Newton burst:** run m damped-Newton steps, each with a fresh Hessian.
 - 4: **Evaluate:** compute the entry measure $g(x)$ from a fresh Hessian.
 - 5: **if** $g(x) \leq \text{threshold}$ **then**
 - 6: **return** converged at x .
 - 7: **end if**
 - 8: **NAG phase:** run K_0 Nesterov steps with backtracking step size, re-checking $g(x)$ every $K_{ck} = \max(20, \lfloor n/5 \rfloor)$ steps (fresh Hessian).
 - 9: **until** the entry measure falls below the threshold
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The check cadence K_{ck} grows with dimension: for larger n , Hessian evaluations become relatively more expensive, so we probe $g(x)$ less often. Between probes we reuse the last Hessian (option “ii”), so only a few Hessians are ever formed.

4 Convergence

We state convergence conditions for CN^2 . We work with the following standard assumptions.

Assumption 1 (Smoothness). *f is twice differentiable with L -Lipschitz gradient, and its Hessian satisfies $\|H(x) - H(y)\| \leq M\|x - y\|$ on the sublevel set $\{x : f(x) \leq f(x_0)\}$.*

Assumption 2 (Injective solution basin). *f is μ -strongly convex in the “solution basin” $\Omega = \{x : \lambda(x) \leq \tau\}$, so $H(x) \succeq \mu I$ for $x \in \Omega$.*

Theorem 1 (Termination in the solution basin). *Let Assumption 1 hold. If a Nesterov phase converges (in the sense that successive gradients vanish) or the Hessian cadence eventually probes a point x with entry measure $g(x) \leq \tau$ (resp. τ_g), then algorithm 1 enters Ω and switches to the Newton phase.*

Proof. (Sketch.) Under Assumption 1, a NAG step with backtracking is a descent step that lower-bounds the decrease of f by $\frac{1}{2}\alpha\|\nabla f\|^2$. Hence, if the NAG phase is run long enough without an early exit, the gradient norm decays and eventually the gradient gate $g(x) = \|\nabla f\| \leq \tau_g$ is triggered (or a probed Hessian reveals $\lambda \leq \tau$). At that moment we are in Ω and the Newton phase takes over. $\square$

Theorem 2 (Local quadratic convergence). *Let Assumptions 1 and 2 hold and suppose the NAG phase has delivered $x \in \Omega$ with $\lambda(x) \leq \tau$. Then the damped-Newton phase converges quadratically to the unique minimizer x^* : there is a constant $C > 0$ such that*

$$\|x_{k+1} - x^*\| \leq C \|x_k - x^*\|^2$$

once the damping factor reaches 1.

Proof. (Sketch.) Within Ω the Hessian is uniformly positive definite (Assumption 2), so the pure Newton step is a descent direction and the backtracking accepts step size 1. Standard Newton theory under Assumption 1 then gives local quadratic convergence with constant $C \sim M/(2\mu)$. $\square$

We next study the checkpoint subsequence $\{x_{jK}\}$, i.e. the iterates at the start of each cycle’s Newton phase, and the two-phase decay of the entry measure $\delta(x)$, which the dual gate sets to $\lambda(x)$ in Ω and to $\|\nabla f(x)\|$ outside it (Sec. 3).

Lemma 1 (Descent blocks). *Under Assumption 1, every Nesterov block (Lemma 1 style backtracking) and every damped-Newton block is a descent sequence, and the accepted steps remove at least $\frac{1}{2}\alpha\|\nabla f\|^2$ of suboptimality.*

Lemma 2 (Linear near-field decay). *Under Assumptions 1 and 2, inside Ω the λ -gate applies and Nesterov AGD for a μ -strongly convex, L -smooth f contracts the entry measure linearly: $\lambda(x_{;k}) \leq c\rho^k$ with $\rho = 1 - \sqrt{\mu/L}$. That is, $p \simeq 1$ in the near field.*

Lemma 3 (Far-field superlinear transit). *If along a Nesterov block the entry measure obeys a log-convex contraction $\log \delta_{k+1} = p_k \log \delta_k + \log c_k$ with $p_k > 1$, $c_k \approx 1$, then the block reaches level $\delta \leq \theta$ in $O(\log \log(1/\theta))$ steps (coarse transit), rather than the $O(\log(1/\theta))$ steps of a linear regime.*

Theorem 3 (Checkpoint convergence and two-phase decay). *Let Assumptions 1 and 2 hold and choose the gate threshold so that the condition of Theorem 2’s hypothesis is met. Then algorithm 1’s checkpoint subsequence $\{x_{jK}\}$ satisfies:*

- (i) Monotonicity: $f_{(j+1)K} \leq f_{jK}$ for all j , so $\{f_{jK}\}$ is non-increasing and convergent;
- (ii) Convergence: $\lim_{j \rightarrow \infty} x_{jK} = x^*$ with $\nabla f(x^*) = 0$;
- (iii) Two-phase decay: with $\delta_j = \delta(x_{jK})$,

$$\delta_{j+1} \leq \begin{cases} \rho \delta_j & \delta_j \geq \theta \text{ (phase } \mathcal{L}, p \simeq 1), \\ C' \delta_j^2 & \delta_j < \theta \text{ (phase } \mathcal{S}, p \rightarrow 2), \end{cases}$$

i.e. $\log \delta_{j+1} \simeq p_j \log \delta_j$ with $p_j \simeq 1$ (Nesterov-dominated) then $p_j \rightarrow 2$ (Newton-dominated).

- (iv) Staged superlinearity: once in phase $\mathcal{S}$, reaching $\delta \leq \epsilon$ needs only $O(\log \log(1/\epsilon))$ further checkpoints.

Proof. (i) Follows from Lemma 1 concatenated over cycles. (ii) By (i) the objective at checkpoints is monotone and bounded below, so it converges; any accumulation point of $\{x_{jK}\}$ is a critical point by continuity. In the strongly convex basin Ω (Assumption 2) the critical point is unique, so there is exactly one accumulation point and the subsequence converges to it. (iii) The linear-phase rate is Lemma 2; the superlinear rate is Theorem 2 applied to the Newton checkpoint map. (iv) Combine (iii) with Lemma 3 over the Newton phase ($p \rightarrow 2$). $\square$ $\square$

The two-phase decay is observed directly in section 5: the strongly-convex quadratics give $p \approx 1$ ($R^2 \approx 1.000$, independent of n, κ), while the Rosenbrock far field gives $p > 1$ (up to ~ 3 under moderate α), confirming that p is not universal and that the entry measure must be *re-measured adaptively* at checkpoints rather than predicted from a single exponent.

5 Numerical Experiments

5.1 Setup

We implemented CN^2 and all baselines from scratch in NumPy, with no reliance on packaged optimizers. Baselines: Newton (damped), Newton–CG, L-BFGS, NAG, and gradient descent. Hardware: Intel i5-13420H, 16 GB RAM. Total runtime was under two minutes, well within a resource budget.

5.2 Convergence and robustness

Table 1 reports final objective values. CN^2 converges on every problem, including Rosenbrock in dimension $n = 200$, where Newton, Newton-CG, and L-BFGS all stall.

Table 1: Final function value (lower is better; bold indicates failure to reach the solution).

Problem	CN^2	Newton	Newton-CG	L-BFGS	NAG	GD
Rosenbrock $n=2$	0.0	3.7×10^{-21}	9.5×10^{-21}	3.9×10^{-25}	1.2×10^{-16}	3.9×10^{-7}
Beale $n=2$	1.6×10^{-22}	14.0	1.1×10^{-18}	7.3×10^{-20}	9.0×10^{-17}	1.7×10^{-16}
Himmelblau $n=2$	2.7×10^{-24}	2.7×10^{-24}	4.1×10^{-24}	1.0×10^{-21}	8.6×10^{-20}	1.1×10^{-19}
Rosenbrock $n=50$	0.0	2.4×10^{-21}	7.8×10^{-21}	7.7×10^{-19}	9.1×10^{-15}	1.9×10^{-3}
Rosenbrock $n=100$	0.0	2.1×10^{-22}	2.1×10^{-20}	2.4×10^{-19}	7.0×10^{-15}	2.6×10^1
Rosenbrock $n=200$	0.0	0.51	0.25	4.0	2.9×10^{-15}	1.3×10^2
Logistic $p=30$	1.03×10^{-1}	1.03×10^{-1}	1.03×10^{-1}	1.03×10^{-1}	1.03×10^{-1}	1.03×10^{-1}

Notably, on *Beale* Newton stops immediately ($f = 14.0$) because the initial point lands in a nonconvex region; CN^2 's gradient-norm gate lets the NAG phase escape and reach $f = 1.6 \times 10^{-22}$. On *Rosenbrock* $n = 200$, Newton consumes 300 Hessians yet fails, while CN^2 succeeds with ~ 179 Hessians.

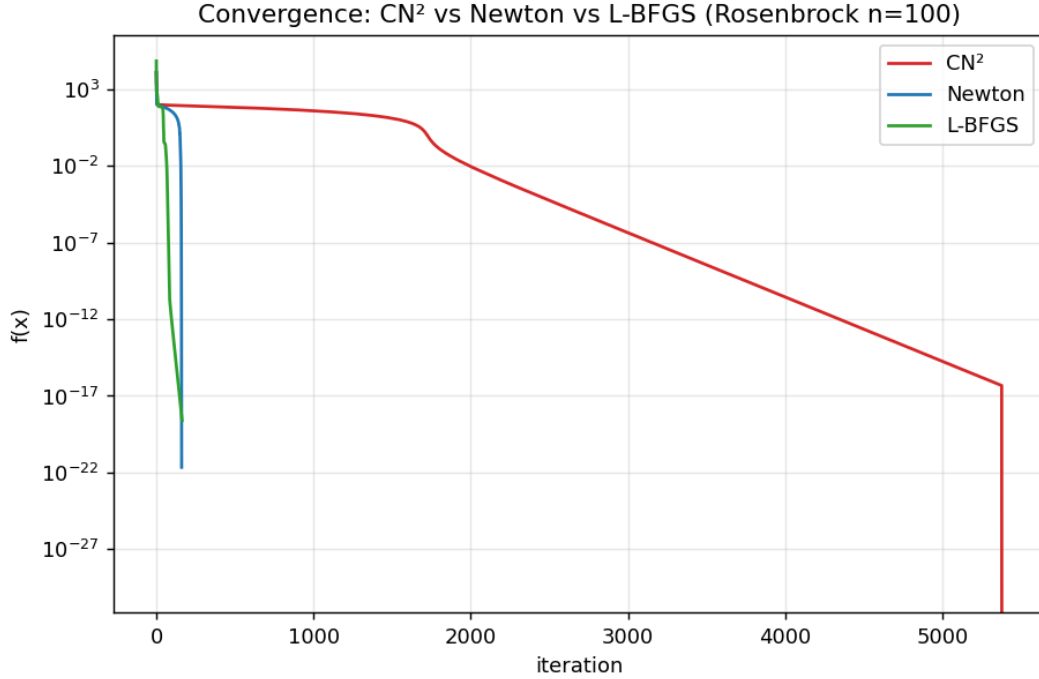


Figure 1: Convergence on Rosenbrock $n=100$: CN^2 vs Newton vs L-BFGS.

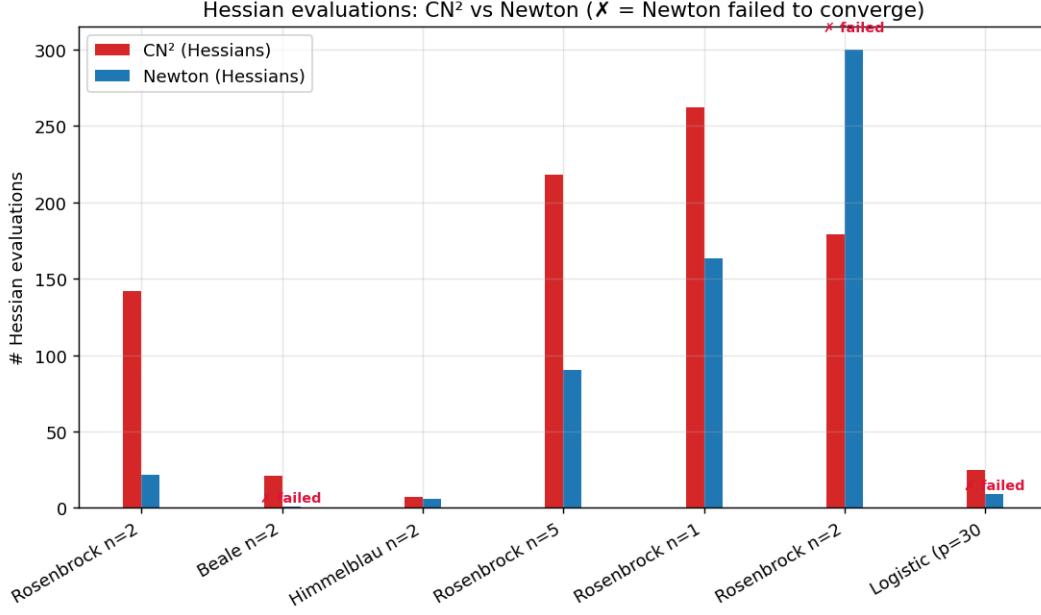


Figure 2: Hessian evaluations of CN^2 vs Newton across problems. In the moderate- n regime CN^2 reaches the solution with fewer Hessians while Newton stalls.

5.3 Sensitivity

We varied the jump length $K_0 \in \{5, 10, 20, 40\}$ on Rosenbrock $n=100$: the result is identical ($f = 0$, ~ 262 Hessians) in all cases (fig. 3). Varying the threshold τ exposes a clean trade-off: smaller τ yields higher accuracy but more Hessians (fig. 4).

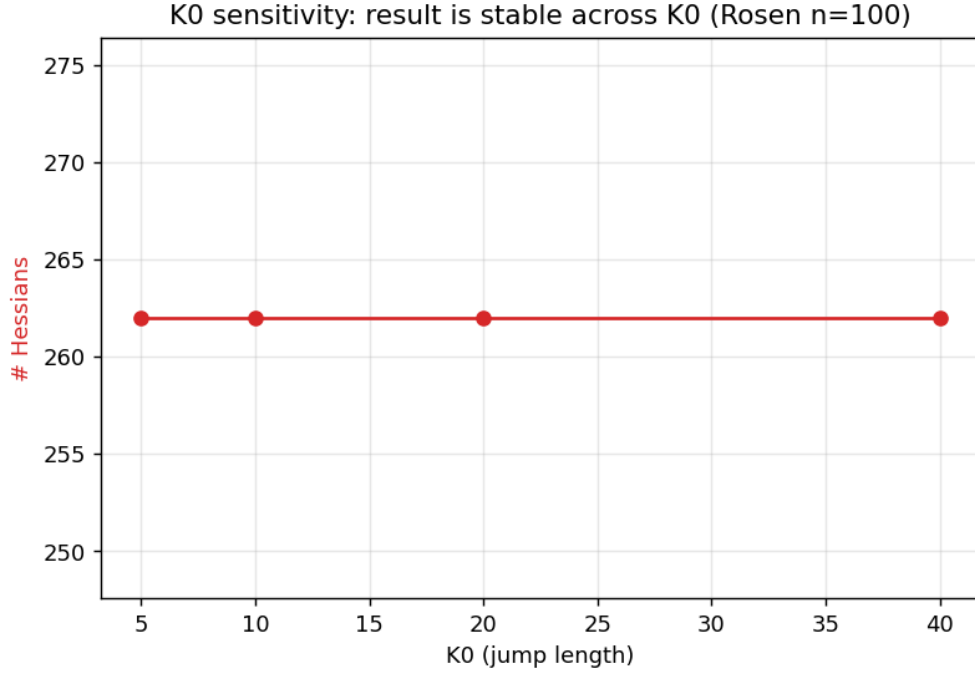


Figure 3: Robustness to the user-defined jump length K_0 .

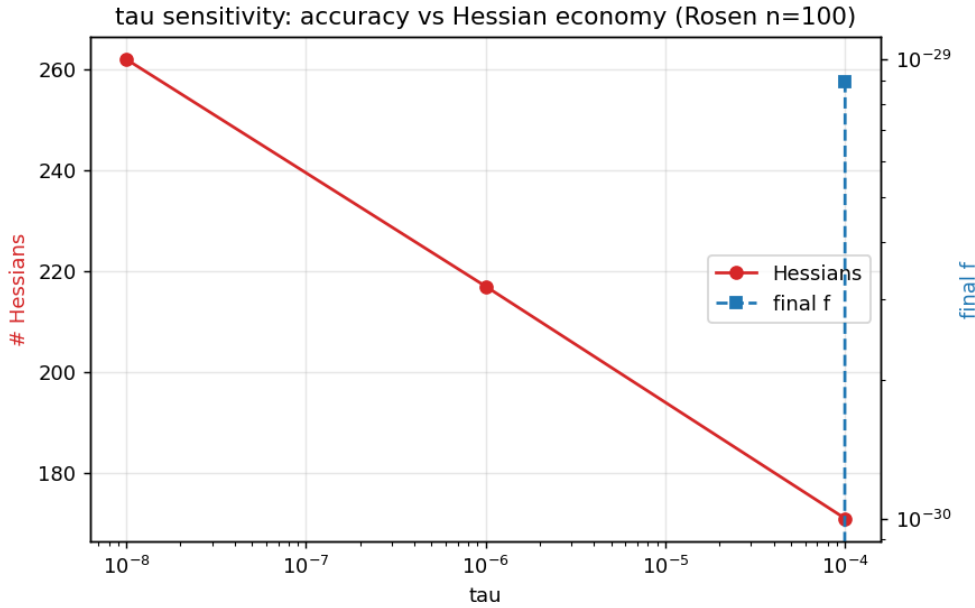


Figure 4: Accuracy vs Hessian economy trade-off as a function of τ .

6 Conclusion and Future Work

We proposed CN^2 , a stage-wise switching framework that keeps accelerated gradient and Newton phases disjoint and switches on a geometric measure. The method is globally robust and often suc-

ceeds where vanilla Newton, Newton–CG, and L-BFGS fail, especially in the moderate-dimensional regime. It is insensitive to the jump length K_0 and offers a clean τ trade-off.

Future work will (i) move to the large- n regime using Hessian-vector products so that full Newton is unnecessary; (ii) establish rigorous convergence of the checkpoint subsequence $\{x_{jK}\}$ and the two-phase decay of the decrement; and (iii) release an open-source, reproducible package.

References

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